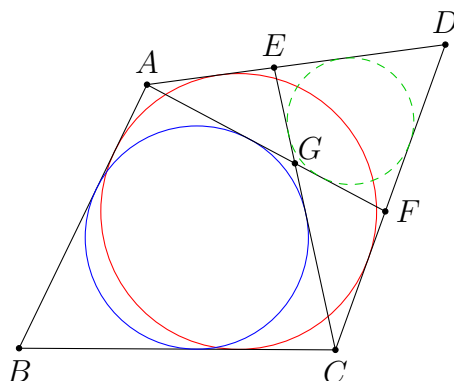


Circumscribed Quadrilaterals — Solutions

Problem 1. Let $ABCD$ be a circumscribed quadrilateral. Let E and F be points on the sides AD and CD respectively. Let G be the intersection point of AF and CE . If $ABCG$ is a circumscribed quadrilateral, prove that $DEGF$ is a circumscribed quadrilateral.



Solution. By Pitot's theorem, we have

$$AB + CD = BC + AD, \quad (1)$$

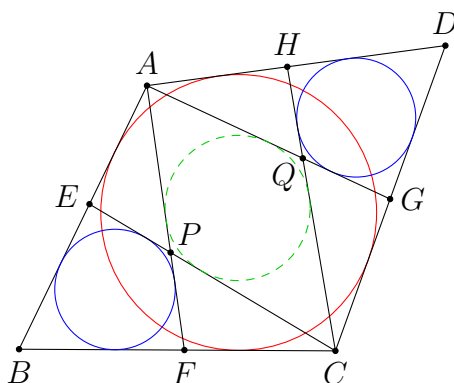
$$AB + CG = BC + AG. \quad (2)$$

Taking the difference of (1) and (2), we obtain

$$CD + AG = AD + CG.$$

This implies the concave quadrilateral $AGCD$ has an incircle, which means $DEGF$ is a circumscribed quadrilateral. $\square$

Problem 2. Let $ABCD$ be a circumscribed quadrilateral. Let E, F, G, H be points on the sides AB, BC, CD, DA respectively. Let P be the intersection point of AF and CE , and let Q be the intersection point of AG and CH . If $BFPE$ and $DHQG$ are circumscribed quadrilaterals, prove that $APCQ$ is a circumscribed quadrilateral.



Solution. Applying Pitot's theorem to $ABCD$, $ABCP$ and $AQCD$, we have

$$AB + CD = BC + AD, \quad (1)$$

$$AB + CP = BC + AP, \quad (2)$$

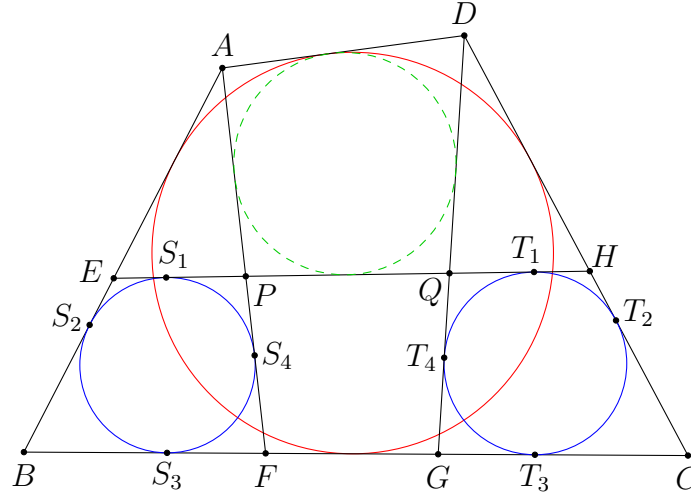
$$AQ + CD = AD + CQ. \quad (3)$$

Computing $(2)+(3)-(1)$, we obtain

$$AQ + CP = AP + CQ.$$

This implies $APCQ$ is a circumscribed quadrilateral. □

Problem 3. Let $ABCD$ be a circumscribed quadrilateral. Let E be a point on the side AB , let F and G be points on the side BC , and let H be a point on the side CD . Let P be the intersection point of AF and EH , and let Q be the intersection point of DG and EH . If $BFPE$ and $CHQG$ are circumscribed quadrilaterals, prove that $APQD$ is a circumscribed quadrilateral.

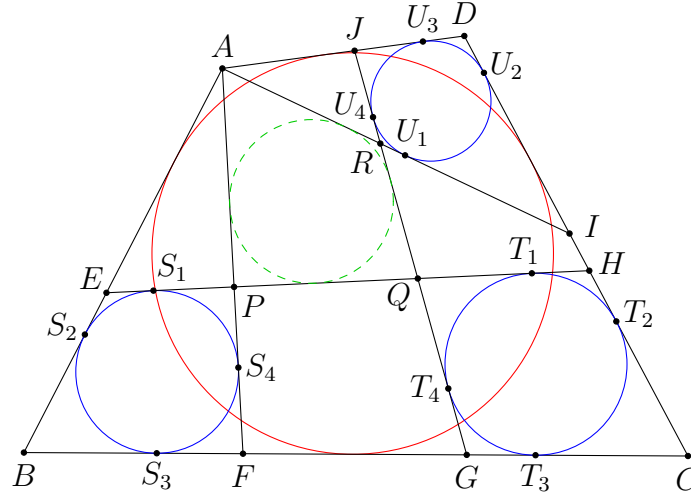


Solution. Let the incircle of $BFPE$ touch EP , BE , BF , FP at S_1 , S_2 , S_3 , S_4 respectively, and let the incircle of $CHQG$ touch HQ , CH , CG , GQ at T_1 , T_2 , T_3 , T_4 respectively. Note that $S_1T_1 = S_3T_3$ as they are symmetric about the line joining the centres of the incircles. By Pitot's theorem, we have $AB + CD = BC + DA$. Now,

$$\begin{aligned} AB + CD &= BC + DA \\ \Rightarrow (AS_2 + BS_2) + (CT_2 + DT_2) &= (BS_3 + S_3T_3 + CT_3) + DA \\ \Rightarrow AS_4 + DT_4 &= S_1T_1 + DA \\ \Rightarrow (AP + PS_4) + (DQ + QT_4) &= (S_1P + PQ + QT_1) + DA \\ \Rightarrow AP + DQ &= PQ + DA. \end{aligned}$$

This implies $APQD$ is a circumscribed quadrilateral. □

Problem 4. Let $ABCD$ be a circumscribed quadrilateral. Let E be a point on the side AB , let F and G be points on the side BC , let H and I be points on the side CD , and let J be a point on the side DA . Let P be the intersection point of AF and EH , let Q be the intersection point of EH and GJ , and let R be the intersection point of GJ and AI . If $BFPE$, $CHQG$ and $DJRI$ are circumscribed quadrilaterals, prove that $APQR$ is a circumscribed quadrilateral.



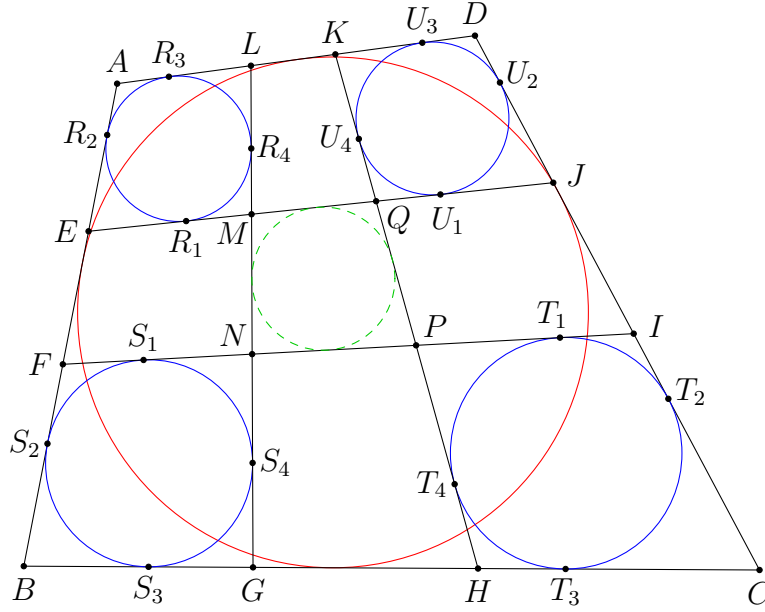
Solution. Let the incircle of $BFPE$ touch EP , BE , BF , FP at S_1 , S_2 , S_3 , S_4 respectively, let the incircle of $CHQG$ touch HQ , CH , CG , GQ at T_1 , T_2 , T_3 , T_4 respectively, and let the incircle of $DJRI$ touch IR , DI , DJ , JR at U_1 , U_2 , U_3 , U_4 respectively. As above, by Pitot's theorem, we have $AB + CD = BC + DA$. Now,

$$\begin{aligned}
 & AB + CD = BC + DA \\
 \Rightarrow & (AS_2 + BS_2) + (CT_2 + T_2U_2 + DU_2) = (BS_3 + S_3T_3 + CT_3) + (AU_3 + DU_3) \\
 \Rightarrow & AS_4 + T_4U_4 = S_1T_1 + AU_1 \\
 \Rightarrow & (AP + PS_4) + (T_4Q + QR + RU_4) = (S_1P + PQ + QT_1) + (AR + RU_1) \\
 \Rightarrow & AP + QR = PQ + AR.
 \end{aligned}$$

This implies $APQR$ is a circumscribed quadrilateral. $\square$

Problem 5. Let $ABCD$ be a circumscribed quadrilateral. Let E and F be points on the side AB , let G and H be points on the side BC , let I and J be points on the side CD , and let K and L be points on the side DA . Let M be the intersection point of GL and EJ , let N be the intersection point of FI and GL , let P be the intersection point of HK and FI , and let Q be the intersection point of EJ and HK . If $AEM L$, $BGN F$, $CIP H$ and $DKQ J$ are circumscribed quadrilaterals, prove that $MNPQ$ is a circumscribed quadrilateral.

Solution. Let the incircle of $AEM L$ touch EM , AE , AL , LM at R_1 , R_2 , R_3 , R_4 respectively, let $BGN F$ touch FN , BF , BG , GN at S_1 , S_2 , S_3 , S_4 respectively, let the incircle of $CIP H$ touch IP , CI , CH , HP at T_1 , T_2 , T_3 , T_4 respectively, and let the incircle of $DKQ J$ touch JQ , DJ , DK , KQ at U_1 , U_2 , U_3 , U_4 respectively.



By Pitot's theorem, we have $AB + CD = BC + DA$. Now,

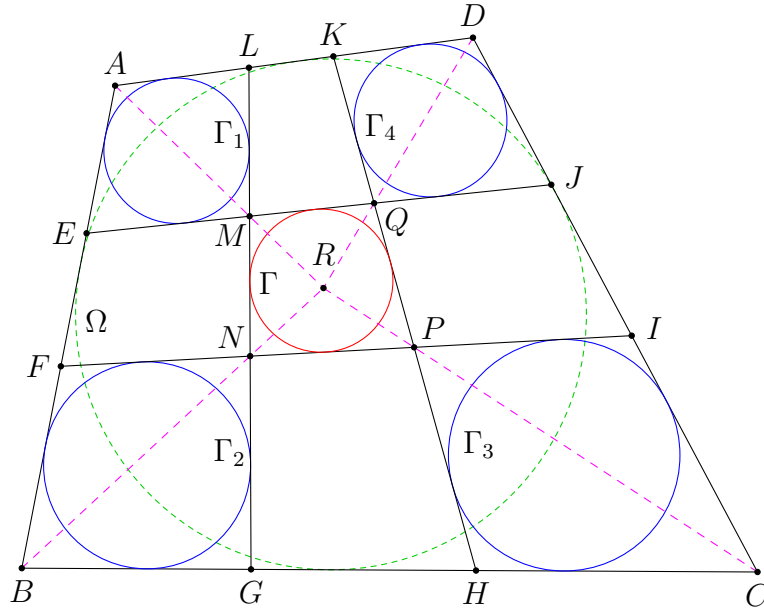
$$\begin{aligned}
 & AB + CD = BC + DA \\
 \Rightarrow & (AR_2 + R_2S_2 + BS_2) + (CT_2 + T_2U_2 + DU_2) = (BS_3 + S_3T_3 + CT_3) + (AR_3 + R_3U_3 + DU_3) \\
 \Rightarrow & R_4S_4 + T_4U_4 = S_1T_1 + R_1U_1 \\
 \Rightarrow & (R_4M + MN + NS_4) + (T_4P + PQ + QU_4) = (S_1N + NP + PT_1) + (R_1M + MQ + QU_1) \\
 \Rightarrow & MN + PQ = NP + MQ.
 \end{aligned}$$

This implies $MNPQ$ is a circumscribed quadrilateral. $\square$

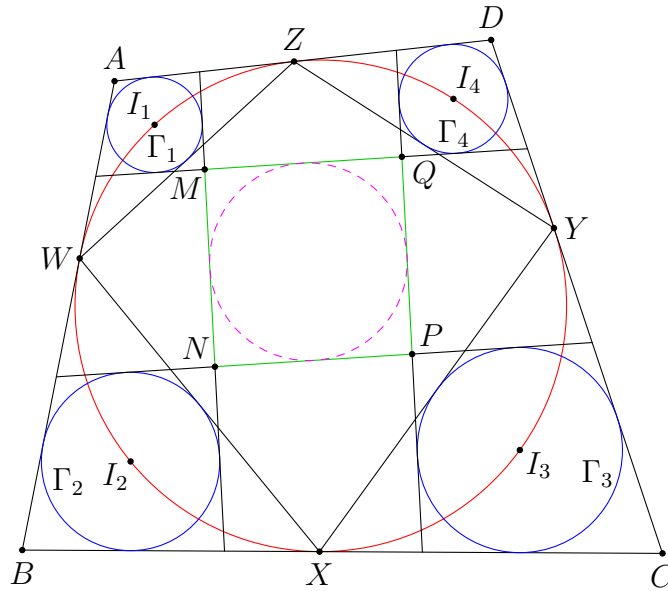
Problem 6. Let $ABCD$ be a quadrilateral. Let E and F be points on the side AB , let G and H be points on the side BC , let I and J be points on the side CD , and let K and L be points on the side DA . Let M be the intersection point of GL and EJ , let N be the intersection point of FI and GL , let P be the intersection point of HK and FI , and let Q be the intersection point of EJ and HK . If $AEML$, $BGNF$, $CIPH$, $DKQJ$ and $MNPQ$ are circumscribed quadrilaterals, prove that AM , BN , CP , DQ are concurrent.

Solution. As in problem 5, we can show that $ABCD$ is a circumscribed quadrilateral. Let $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4, \Gamma, \Omega$ be the incircles of $AEML, BGNF, CIPH, DKQJ, MNPQ, ABCD$ respectively. Let R be the internal similitude centre of Γ and Ω .

Applying Monge's theorem to Γ_1, Γ, Ω , we see that the external similitude centre A of Γ_1 and Ω , the internal similitude centre M of Γ_1 and Γ , and the internal similitude centre R of Γ and Ω are collinear. Similarly, all of BN, CP, DQ pass through R . $\square$



Problem 8. Let $ABCD$ be a circumscribed quadrilateral. Let W, X, Y, Z be the tangential points of the incircle of $ABCD$ with AB, BC, CD, DA respectively. Let $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ be the incircles of $\triangle AZW, \triangle BWX, \triangle CXY, \triangle DYZ$ respectively. The external common tangent of Γ_1 and Γ_2 other than AB , the external common tangent of Γ_2 and Γ_3 other than BC , the external common tangent of Γ_3 and Γ_4 other than CD , and the external common tangent of Γ_4 and Γ_1 other than DA enclose a quadrilateral $MNPQ$. Prove that $MNPQ$ is a rhombus.



Solution. We first prove that the centres I_1, I_2, I_3, I_4 of $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ lie on the incircle of $ABCD$. Indeed, note that $AW = AZ$, and so $\angle AWI_1 = \angle ZWI_1 = \angle WZI_1 = \angle AZI_1$. This shows

(WZI_1) is tangent to AB and AD at W and Z respectively, and hence (WZI_1) is the incircle of $ABCD$. Similarly, all of I_2, I_3, I_4 lie on this incircle.

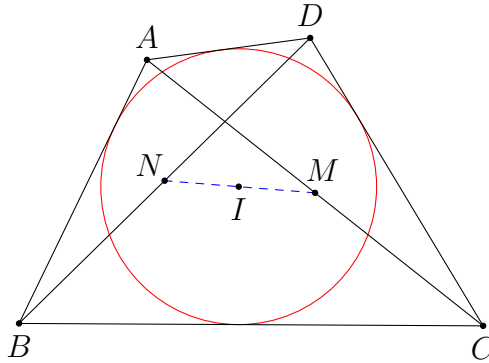
Next, we prove that $MNPQ$ is a parallelogram by showing the opposite sides are parallel. We have

$$\begin{aligned}
\angle(MN, PQ) &= \angle(MN, AB) + \angle(AB, CD) + \angle(CD, PQ) \\
&= 2\angle(I_1I_2, AB) + \angle(AB, CD) + 2\angle(CD, I_3I_4) \\
&= 2(\angle I_2WB - \angle WI_2I_1) + \angle(AB, CD) + 2(\angle CYI_3 - \angle I_4I_3Y) \\
&= 2\angle I_2WB - 2\angle WZI_1 + \angle(AB, CD) + 2\angle CYI_3 - 2\angle I_4ZY \\
&= \angle XWB - \angle WZA + \angle(AB, CD) + \angle CYX - \angle DZY \\
&= \angle(AB, CD) - \angle BWX + \angle AZW - \angle XYC + \angle YZD \\
&= (180^\circ - B - C) - \left(90^\circ - \frac{B}{2}\right) + \left(90^\circ - \frac{A}{2}\right) - \left(90^\circ - \frac{C}{2}\right) + \left(90^\circ - \frac{D}{2}\right) \\
&= 180^\circ - \frac{A + B + C + D}{2} = 0^\circ.
\end{aligned}$$

This proves $MN \parallel PQ$. Similarly, $MQ \parallel NP$. So $MNPQ$ is a parallelogram.

Lastly, by problem 5, $MNPQ$ is a circumscribed quadrilateral. This yields $MN + PQ = MQ + NP$, and hence $MN = PQ = MQ = NP$. This means $MNPQ$ is a rhombus. $\square$

Problem 9. (Newton's theorem) Let $ABCD$ be a circumscribed quadrilateral with incentre I . Let M and N be the midpoints of AC and BD respectively. Prove that M, I, N are collinear.



Solution. Consider the locus of all points X inside $ABCD$ satisfying

$$[XAB] + [XCD] = [XAD] + [XBC].$$

Using the coordinate system, this is a linear equation. (For example, if $(x, y), (a, b), (c, d)$ are the coordinates of X, A, B respectively, then $[XAB] = \frac{1}{2}[(b - d)x + (c - a)y + (ad - bc)]$ is a linear equation in x, y .) Therefore, the locus is either the entire quadrilateral $ABCD$, a segment, or the empty set.

Note that $[NAB] = [NAD]$ and $[NCD] = [NBC]$. So N is a possible choice of X . Similarly, M is a possible choice of X . In addition, let r be the inradius of $ABCD$. Then

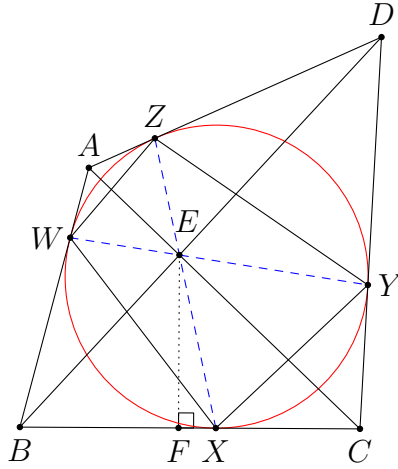
$$[IAB] + [ICD] = \frac{AB}{2}r + \frac{CD}{2}r = \frac{AD}{2}r + \frac{BC}{2}r = [IAD] + [IBC].$$

So I is also a possible choice of X . We are done unless the locus is the entire quadrilateral $ABCD$.

If $[XAB] + [XCD] = [XAD] + [XBC]$ is an identity, we consider any point X on the side AB . This gives $[XCD] = [XAD] + [XBC]$, and hence $[XCD] = \frac{1}{2}[ABCD]$. This shows $[XCD]$ is a constant, and hence $AB \parallel CD$. Similarly, $AD \parallel BC$. Thus, $ABCD$ is a parallelogram (indeed it is a rhombus). In that case $M = N$, and so M, I, N are still collinear. $\square$

Problem 10. Let $ABCD$ be a circumscribed quadrilateral. Let E be the intersection point of AC and BD . Let a, b, c, d be the distances from E to AB, BC, CD, DA respectively. Prove that

$$\frac{1}{a} + \frac{1}{c} = \frac{1}{b} + \frac{1}{d}.$$



Solution. Let W, X, Y, Z be the tangential points of the incircle of $ABCD$ with AB, BC, CD, DA respectively, and let F be the projection of E onto BC . Applying Pascal's theorem to $WWYZZX$ and $YYWXXZ$, we know that the four points $WW \cap ZZ = A$, $WY \cap XZ$, $WX \cap YZ$ and $YY \cap XX = C$ are collinear. In particular, $WY \cap XZ$ lies on AC . Similarly, $WY \cap XZ$ lies on BD , and so $WY \cap XZ = E$.

Now we have

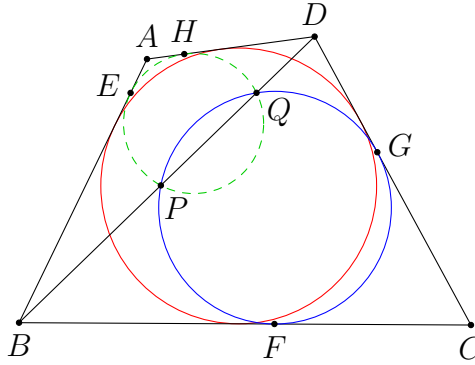
$$b = EF = EX \sin \angle EXF = EX \sin \angle ZYX.$$

Similarly, we find that $a = EW \sin \angle YXW$, $c = EY \sin \angle YXW$ and $d = EZ \sin \angle ZYX$. Therefore,

$$\begin{aligned} \frac{1}{a} + \frac{1}{c} &= \frac{1}{b} + \frac{1}{d} \\ \Leftrightarrow \frac{EW + EY}{EW \times EY \sin \angle YXW} &= \frac{EX + EZ}{EX \times EZ \sin \angle ZYX} \\ \Leftrightarrow \frac{WY}{\sin \angle YXW} &= \frac{XZ}{\sin \angle ZYX}. \end{aligned}$$

This is true by the extended sine law. □

Problem 11. Let $ABCD$ be a circumscribed quadrilateral. A circle tangent to BC and CD at F and G intersects the diagonal BD at P and Q . Prove that there exists a circle passing through P and Q which is tangent to AB and AD .

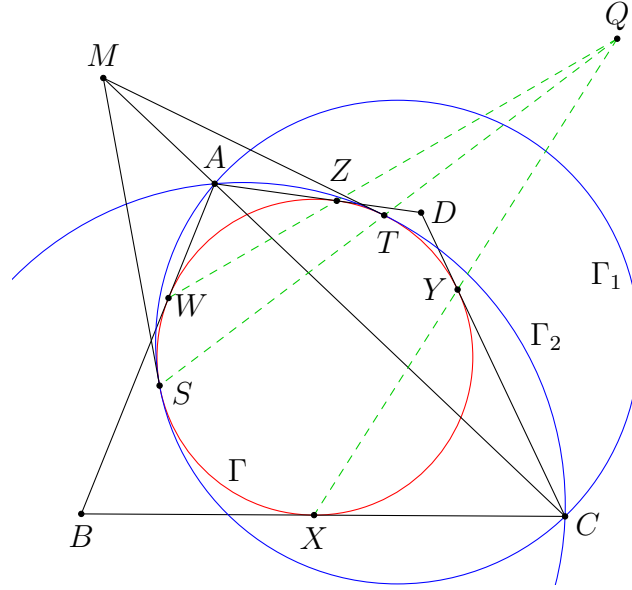


Solution. Note that there exists a circle tangent to AB and AD at E and H which also passes through Q . Suppose this circle meets BD again at P' . It suffices to prove $P' = P$. Using the relations $AB + CD = BC + AD$, $AE = AH$ and $CF = CG$, we obtain $BE + DG = BF + DH$. Using the power of point theorem, this becomes

$$\sqrt{BP' \times BQ} + \sqrt{DP \times DQ} = \sqrt{BP \times BQ} + \sqrt{DP' \times DQ}.$$

If P' lies between B and P , then $BP' < BP$ and $DP < DP'$, which is impossible as the left-hand side is smaller. Similarly, if P lies between B and P' , then $BP' > BP$ and $DP > DP'$, which is also impossible. Therefore, we must have $P' = P$ as desired. □

Problem 14. Let $ABCD$ be a circumscribed quadrilateral with incircle Γ . Let W, X, Y, Z be the tangential points of Γ with AB, BC, CD, DA respectively. There are two circles Γ_1 and Γ_2 passing through A and C , and tangent to Γ at S and T respectively. Prove that WZ, ST, XY are concurrent.

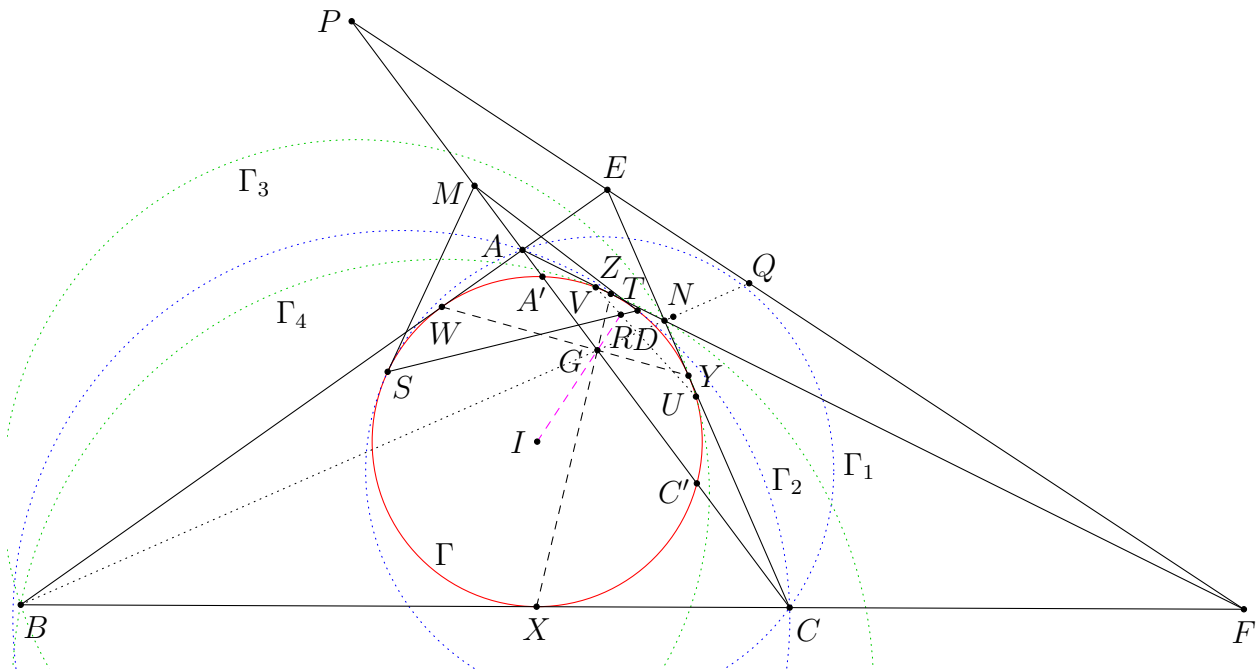


Solution. By considering the radical centre of $\Gamma, \Gamma_1, \Gamma_2$, we see that the tangents at S and T to Γ meet at a point M lying on the line AC . Since the tangents from A to Γ touch Γ at W and Z , the line WZ is the polar of A w.r.t. Γ . Similarly, XY is the polar of C . Let Q be the intersection point of WZ and XY . Then the polar of Q is AC by La Hire's theorem. Since this polar passes through M , the polar of M passes through Q by La Hire's theorem. But the polar of M is ST . Therefore, S, T, Q are collinear, which means WZ, ST, XY are concurrent at Q . $\square$

Problem 15. Let $ABCD$ be a circumscribed quadrilateral with incircle Γ and incentre I . Let AC meet BD at G . There are two circles Γ_1 and Γ_2 passing through A and C , and tangent to Γ at S and T respectively. There are two circles Γ_3 and Γ_4 passing through B and D , and tangent to Γ at U and V respectively. Let R be the intersection point of ST and UV . Prove that I, G, R are collinear.

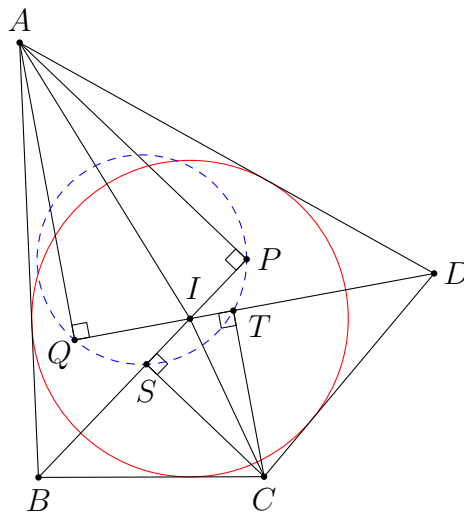
Solution. Let W, X, Y, Z be the tangential points of Γ with AB, BC, CD, DA respectively. Recall that WY and XZ intersect at G (see the proof of problem 10). Let E be the intersection point of AB and CD , and let F be the intersection point of AD and BC . Let P and Q be the intersection points of EF with AC and BD respectively. Let M and N be the midpoints of GP and GQ respectively. In addition, suppose Γ meets AC at A' and C' .

We first prove that the tangents to Γ at S and T meet at M . Indeed, suppose they meet at M' . Note that M' lies on AC as it is the radical centre of $\Gamma, \Gamma_1, \Gamma_2$. Using the power of point theorem, we have $M'A \times M'C = M'S^2 = M'A' \times M'C'$. Thus, it suffices to prove $MA \times MC = MA' \times MC'$. Note that both $(P, G; A, C)$ and $(P, G; A', C')$ are harmonic divisions. Indeed, the former holds since the cevians BQ, EC, FA in $\triangle BEF$ intersect at D , which gives $(BP, BG; BA, BC) = -1$, while the latter holds since the polar of P w.r.t. Γ passes through G since the polar of G is EF . Therefore, the midpoint M of PG satisfies $MG^2 = MA \times MC$ and $MG^2 = MA' \times MC'$. This proves $MA \times MC = MA' \times MC'$ as desired. Therefore, the tangents to Γ at S and T meet at M .



It follows that the polar of M w.r.t. Γ is ST . By symmetry, the polar of N is UV . As R is the intersection of ST and UV , this implies the polar of R is MN . Also, the polar of G is EF , or just PQ , which is parallel to MN by the midpoint theorem. As I is the centre of Γ , this implies I, G, R are collinear. This completes the proof. $\square$

Problem 16. Let $ABCD$ be a circumscribed quadrilateral with incentre I . Let P and Q be the projections of A onto the lines BI and DI respectively. Let S and T be the projections of C onto the lines BI and DI respectively. Prove that P, Q, S, T are concyclic.



Solution. We have

$$\frac{AP}{AQ} = \frac{AI \sin \angle AIP}{AI \sin \angle AIQ} = \frac{\sin \angle AIB}{\sin \angle AID} = \frac{\sin \frac{A+B}{2}}{\sin \frac{A+D}{2}}.$$

By symmetry, we have

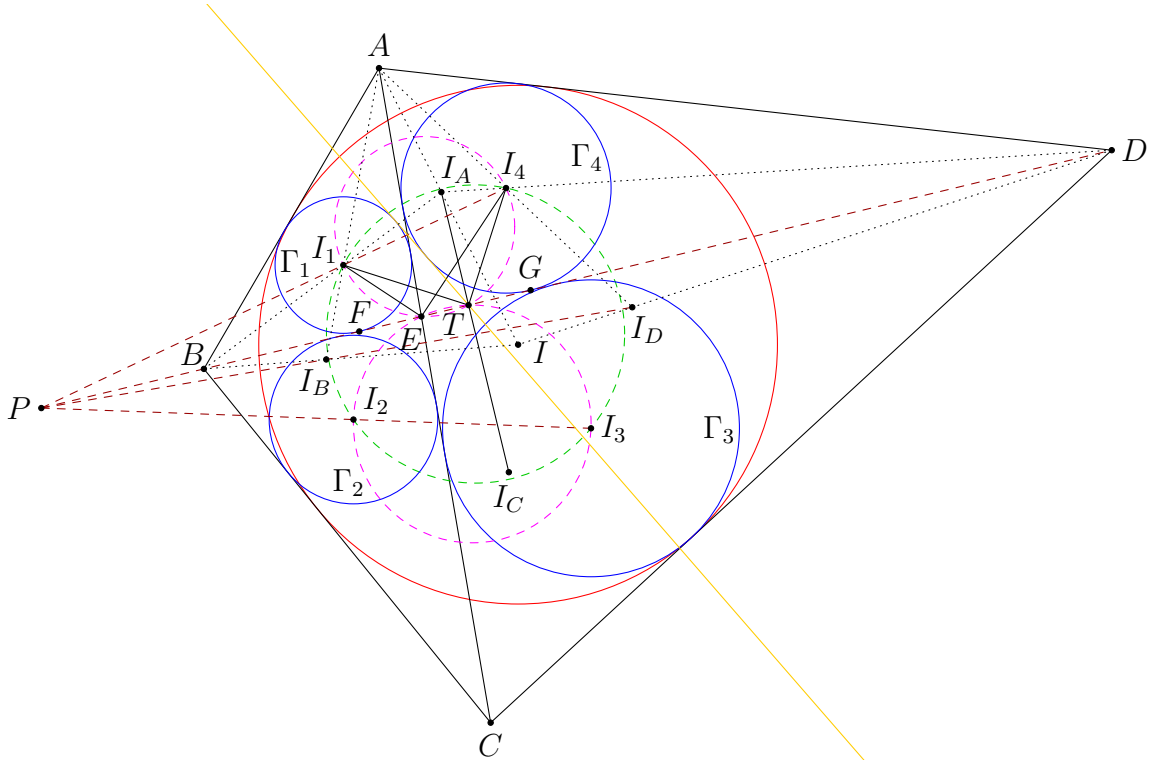
$$\frac{CT}{CS} = \frac{\sin \frac{C+D}{2}}{\sin \frac{C+B}{2}} = \frac{\sin \frac{A+B}{2}}{\sin \frac{A+D}{2}} = \frac{AP}{AQ}$$

since $\frac{C+D}{2} + \frac{A+B}{2} = \frac{C+B}{2} + \frac{A+D}{2} = 180^\circ$. Together with

$$\angle PAQ = 180^\circ - \angle PIQ = 180^\circ - \angle SIT = \angle TCS,$$

we obtain $\triangle APQ \sim \triangle CTS$. Since $\angle IPA = 90^\circ = \angle ITC$ and $\angle IQA = 90^\circ = \angle ISC$, the point I corresponds to itself under the similarity. In other words, $APIQ$ is similar to $CTIS$. Therefore, we have $\angle IPQ = \angle ITS$, which implies P, Q, S, T are concyclic. $\square$

Problem 17. Let $ABCD$ be a circumscribed quadrilateral with incentre I . Let AC meet BD at E . Let I_1, I_2, I_3, I_4 be the incentres of $\triangle ABE, \triangle BCE, \triangle CDE, \triangle DAE$ respectively. Prove that I_1, I_2, I_3, I_4 are concyclic.



Solution. Let $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ be the incircles of $\triangle ABE, \triangle BCE, \triangle CDE, \triangle DAE$ respectively. Let I_A, I_B, I_C, I_D be the incentres of $\triangle ABD, \triangle BCA, \triangle CDB, \triangle DAC$ respectively. We first prove the following well-known results.

Claim 1: $I_A I_C \perp BD$

Proof. Let T_1 and T_2 be the contact points of the incircles of $\triangle ABD$ and $\triangle CBD$ with the side BD respectively. Then we have

$$BT_1 = \frac{AB + BD - AD}{2} = \frac{CB + BD - CD}{2} = BT_2$$

since $AB + CD = BC + AD$. This implies $T_1 = T_2$, and so $I_A I_C \perp BD$. $\square$

Let T be the intersection point of $I_A I_C$ and BD .

Claim 2: T lies on an internal common tangent of Γ_1 and Γ_4 .

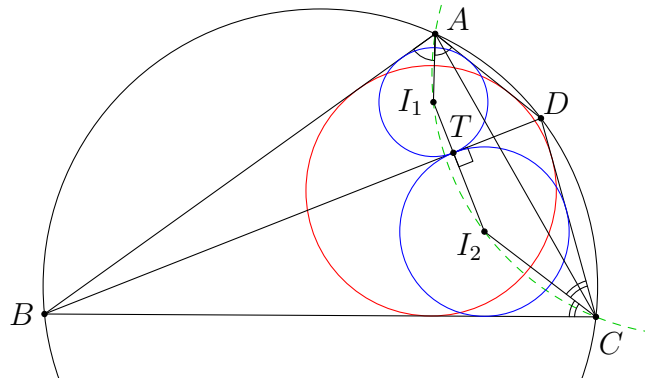
Proof. Let F and G be the contact points of Γ_1 and Γ_4 with BD respectively. We have

$$FT = BT - BF = \frac{AB + BD - AD}{2} - \frac{AB + BE - AE}{2} = \frac{DE + AE - AD}{2} = EG.$$

Since E lies on an internal common tangent of Γ_1 and Γ_4 , and F and G are the contact points of an external common tangent, T lies on the other internal common tangent. $\square$

As E and T are the intersection points of the internal common tangents and an external common tangent of Γ_1 and Γ_4 , we have $\angle I_1 E I_4 = \angle I_1 T I_4 = 90^\circ$. Therefore, I_1, E, T, I_4 are concyclic. Consider $\triangle BI_B I_1$ and $\triangle DI_D I_4$. Their corresponding sidelines meet at the points $BI_B \cap DI_D = I$, $BI_1 \cap DI_4 = I_A$ and $I_B I_1 \cap I_D I_4 = A$ by using the angle bisectors. As these points are collinear, the two triangles are perspective. By Desargues' theorem, $BD, I_B I_D, I_1 I_4$ are concurrent at a point P . By symmetry, I_2, E, T, I_3 are concyclic, and $BD, I_B I_D, I_2 I_3$ are concurrent. This shows all of $BD, I_B I_D, I_1 I_4, I_2 I_3$ pass through P . Since $PI_1 \times PI_4 = PE \times PT = PI_2 \times PI_3$, we see that I_1, I_2, I_3, I_4 are concyclic. $\square$

Problem 19. Let $ABCD$ be a bicentric quadrilateral. Let I_1 and I_2 be the incentres of $\triangle ABD$ and $\triangle BCD$ respectively. Prove that C, A, I_1, I_2 are concyclic.

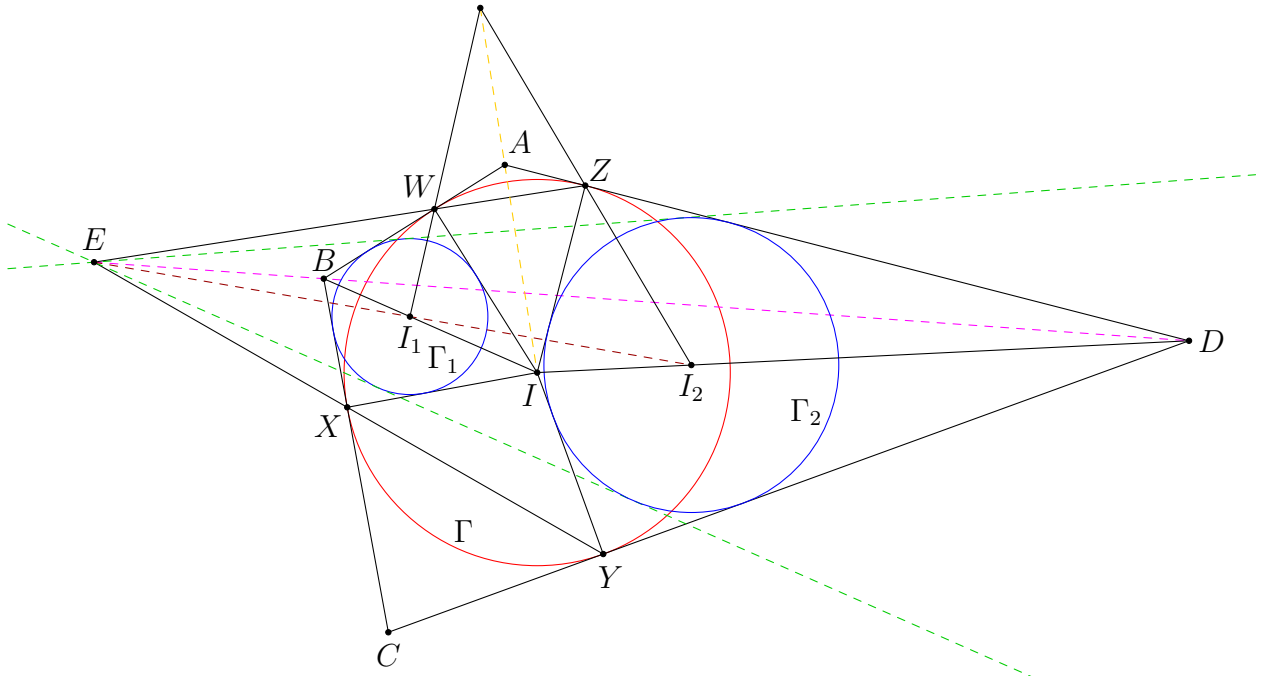


Solution. Recall from the proof of problem 17 that $I_1I_2 \perp BD$. Let T be the intersection point of I_1I_2 and BD . Then we have

$$\begin{aligned}\angle I_1AC + \angle I_1I_2C &= \left(\frac{A}{2} - \angle CAD\right) + \left(360^\circ - 90^\circ - \angle BDC - \frac{C}{2}\right) \\ &= \frac{180 - C}{2} - \angle CBD + 270^\circ - \angle BDC - \frac{C}{2} \\ &= 360^\circ - C - \angle CBD - \angle BDC \\ &= 180^\circ.\end{aligned}$$

This proves C, A, I_1, I_2 are concyclic. □

Problem 20. Let $ABCD$ be a circumscribed quadrilateral with incircle Γ and incentre I . Let W, X, Y, Z be the tangential points of Γ with AB, BC, CD, DA respectively. Let E be the intersection point of WZ and XY . Let Γ_1 and Γ_2 be the incircles of $BXIW$ and $DZIY$ respectively. Prove that E is the intersection of the external common tangents of Γ_1 and Γ_2 .

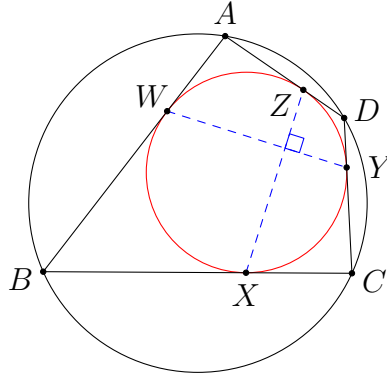


Solution. Let I_1 and I_2 be the centres of Γ_1 and Γ_2 respectively. Note that B, I_1, I are collinear, D, I_2, I are collinear. Also, WI_1, ZI_2, AI are concurrent by symmetry, since AI is the axis of symmetry of $AWIZ$, and WI_1 and ZI_2 are external angle bisectors of $\angle AWI$ and $\angle AZI$ respectively. Therefore, $\triangle BWI_1$ and $\triangle DZI_2$ are perspective. By Desargues' theorem, BD, WZ, I_1I_2 are concurrent. Moreover, E lies on BD as in the proof of problem 10. This proves E is the intersection point of BD and I_1I_2 .

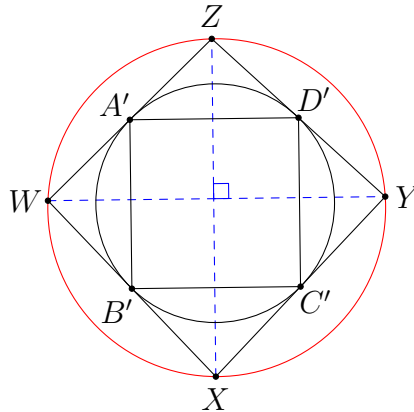
Now, note that B is the external similitude centre of Γ and Γ_1 , while D is the external similitude centre of Γ and Γ_2 . By Monge's theorem, the external similitude centre of Γ_1 and Γ_2 must lie on

BD . It also lies on the line I_1I_2 joining the centres of Γ_1 and Γ_2 . Thus, it is simply the point E as desired. $\square$

Problem 21. Let $ABCD$ be a bicentric quadrilateral. Let W, X, Y, Z be the tangential points of the incircle of $ABCD$ with AB, BC, CD, DA respectively. Prove that $WY \perp XZ$.

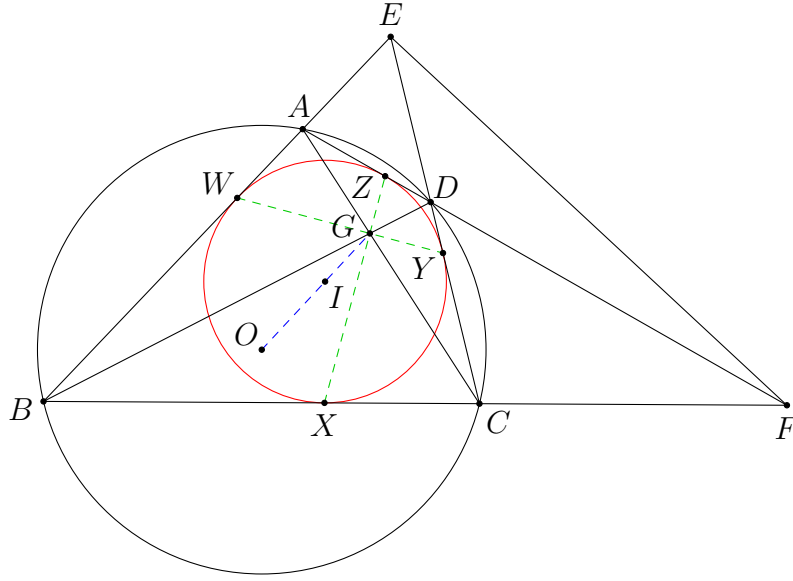


Solution. Consider the inversion w.r.t. the incircle of $ABCD$. Let A', B', C', D' be the images of A, B, C, D respectively. Note that they are just the midpoints of ZW, WX, XY, YZ respectively. By the midpoint theorem, we have $A'B' \parallel ZX \parallel D'C'$ and similarly $A'D' \parallel WY \parallel B'C'$. Therefore, $A'B'C'D'$ is a parallelogram. Since $ABCD$ is cyclic, so is $A'B'C'D'$. So $A'B'C'D'$ is a rectangle. This proves $A'D' \perp A'B'$, and hence $WY \perp XZ$. $\square$



Problem 22. Let $ABCD$ be a bicentric quadrilateral with incentre I and circumcentre O . Let G be the intersection point of AC and BD . Prove that O, I, G are collinear.

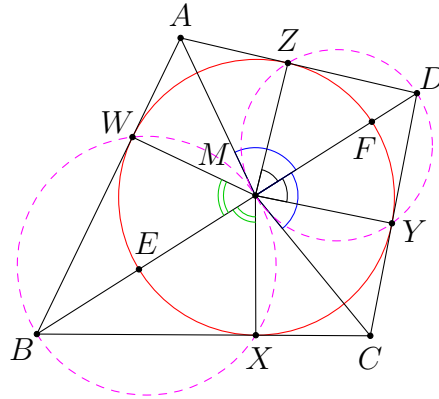
Solution. Let W, X, Y, Z be the tangential points of the incircle of $ABCD$ with AB, BC, CD, DA respectively. Let E be the intersection point of AB and CD . Let F be the intersection point of AD and BC . By the proof of problem 10, G lies on WY and XZ .



Since the polar of E w.r.t. the incircle is WY , the polar of G passes through E by La Hire's theorem. Similarly, the polar of G passes through F . Therefore, EF is the polar of G . This implies $IG \perp EF$.

Next, it is well-known that the polar of G w.r.t. the circumcircle is EF . Therefore, we also have $OG \perp EF$. Combining these, we see that O, I, G are collinear. $\square$

Problem 23. Let $ABCD$ be a circumscribed quadrilateral. Let E and F be the intersection points of BD with the incircle of $ABCD$. Let M be the midpoint of EF . Prove that $\angle AMD = \angle CMD$.



Solution. Let W, X, Y, Z be the tangential points of the incircle with AB, BC, CD, DA respectively. By problem 24 below, we have $\angle DMZ = \angle DMY$ and $\angle BMW = \angle BMX$. Also, by its proof, we know that B, X, M, W are concyclic, D, Z, M, Y are concyclic. Using the sine law, we have

$$\frac{\sin \angle AZM}{\sin \angle AMZ} = \frac{AM}{AZ} = \frac{AM}{AW} = \frac{\sin \angle AWM}{\sin \angle AMW}.$$

Similarly, we have $\frac{\sin \angle CYM}{\sin \angle CMY} = \frac{\sin \angle CXM}{\sin \angle CMX}$. Note that $\angle AWM = \angle BXM = 180^\circ - \angle CXM$ and similarly $\angle AZM = 180^\circ - \angle CYM$. Therefore, we obtain

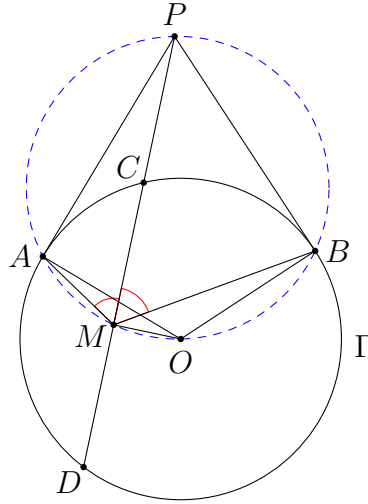
$$\frac{\sin \angle AMW}{\sin \angle AMZ} = \frac{\sin \angle AWM}{\sin \angle AZM} = \frac{\sin \angle CXM}{\sin \angle CYM} = \frac{\sin \angle CMX}{\sin \angle CMY}.$$

As $\angle AMW + \angle AMZ = 180^\circ - \angle DMZ - \angle BMW = \angle CMX + \angle CMY$, and the sum is between 0° and 180° , we have $\angle AMW = \angle CMX$ and $\angle AMZ = \angle CMY$. This implies

$$\angle AMD = \angle AMZ + \angle DMZ = \angle CMY + \angle DMY = \angle CMD.$$

□

Problem 24. Let P be a point lying outside a circle Γ . The tangents from P to Γ touch Γ at A and B . A line passing through P meets Γ at C and D . Let M be the midpoint of CD . Prove that $\angle AMP = \angle BMP$.

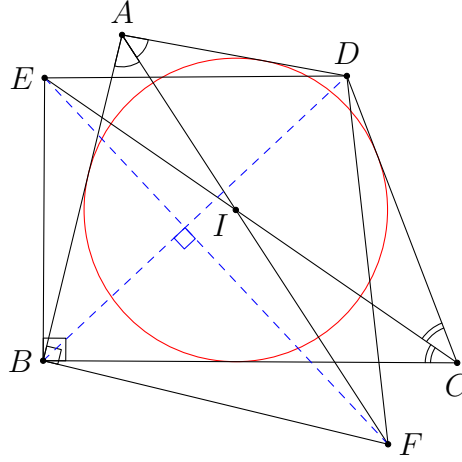


Solution. Let O be the centre of Γ . Since M is the midpoint of CD , we have $\angle OMP = 90^\circ$. Also, we have $\angle OAP = \angle OBP = 90^\circ$. Therefore, P, A, M, O, B are concyclic. As $PA = PB$, the opposite angles $\angle PMA$ and $\angle PMB$ are equal. □

Problem 25. Let $ABCD$ be a circumscribed quadrilateral with incentre I . Let E be the point on the line CI such that $BE \perp BC$. Let F be the point on AI such that $BA \perp BF$. Prove that $BD \perp EF$.

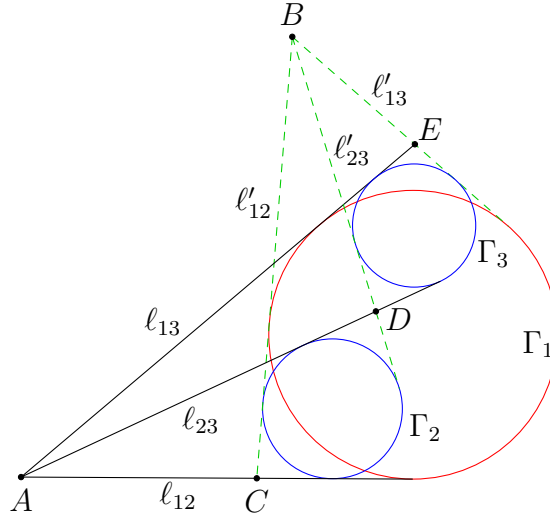
Solution. We have

$$\begin{aligned} DE^2 - BE^2 &= CD^2 + CE^2 - 2CD \times CE \cos \angle DCE - BE^2 \\ &= CD^2 + (BC^2 + BE^2) - 2CD \times CE \cos \angle ECB - BE^2 \\ &= CD^2 + BC^2 - 2CD \times BC \\ &= (CD - BC)^2. \end{aligned}$$



Similarly, we have $DF^2 - BF^2 = (AD - AB)^2$. As $CD - BC = AD - AB$, this implies $DE^2 - BE^2 = DF^2 - BF^2$, and hence $BD \perp EF$. $\square$

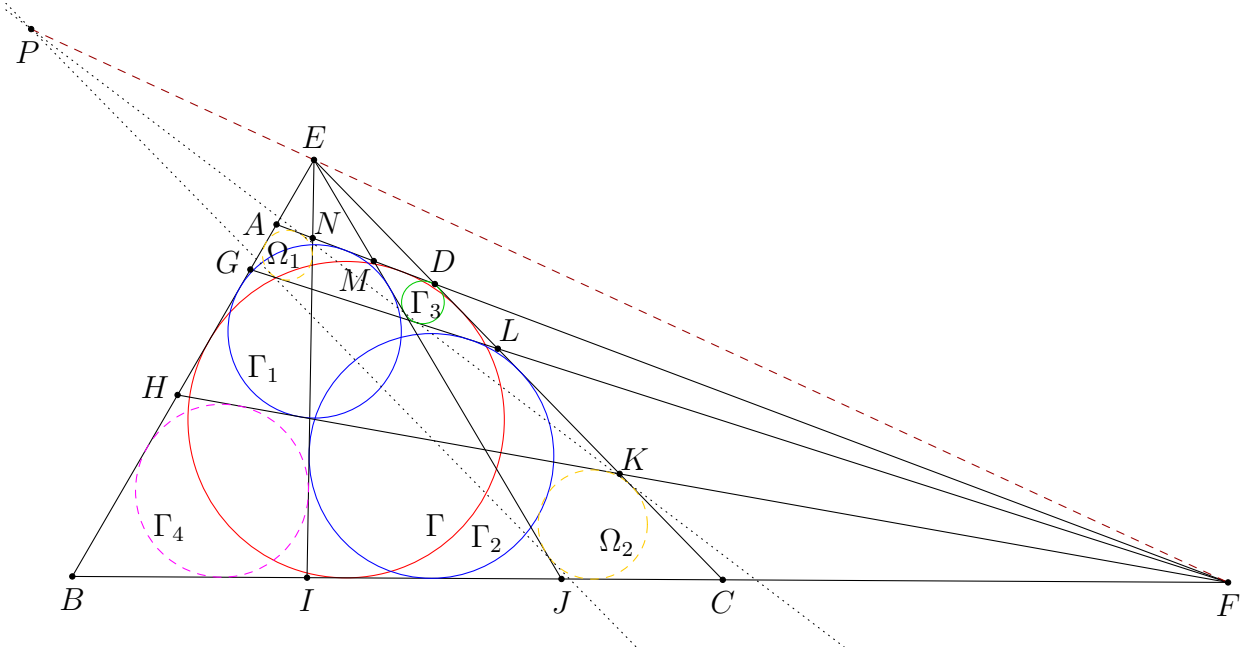
Problem 26. Let $\Gamma_1, \Gamma_2, \Gamma_3$ be three circles. Let ℓ_{12} and ℓ'_{12} be the external common tangents of Γ_1 and Γ_2 . Let ℓ_{13} and ℓ'_{13} be the external common tangents of Γ_1 and Γ_3 . Let ℓ_{23} and ℓ'_{23} be the internal common tangents of Γ_2 and Γ_3 . If $\ell_{12}, \ell_{13}, \ell_{23}$ are concurrent at a point A , prove that $\ell'_{12}, \ell'_{13}, \ell'_{23}$ are concurrent.



Solution. Suppose ℓ'_{12} meets ℓ'_{23} at B . We redefine ℓ'_{13} as the tangent from B to Γ_3 which is different from ℓ'_{23} . It suffices to show that ℓ'_{13} is tangent to Γ_1 . Equivalently, we have to show that $\ell'_{13}, \ell'_{12}, \ell_{13}, \ell_{12}$ enclose an incircle.

Let C be the intersection point of ℓ_{12} and ℓ'_{12} . Let D be the intersection point of ℓ_{23} and ℓ'_{23} . Let E be the intersection point of ℓ_{13} and ℓ'_{13} . Since $\ell_{12}, \ell_{23}, \ell'_{12}, \ell'_{23}$ enclose the incircle Γ_2 , we have $AC + BC = AD + BD$ as analogous to Pitot's theorem. Similarly, we have $AD + BD = AE + BE$. Therefore, we obtain $AC + BC = AE + BE$. This implies $\ell'_{13}, \ell'_{12}, \ell_{13}, \ell_{12}$ enclose an incircle, and so we are done. $\square$

Problem 27. Let $ABCD$ be a circumscribed quadrilateral with incircle Γ . Let G and H be points on the side AB , let I and J be points on the side BC , let K and L be points on the side CD , and let M and N be points on the side DA . Suppose AB, NI, MJ, CD are concurrent at a point E , and AD, GL, HK, BC are concurrent at a point F . The lines BE, EJ, AF, FH enclose an incircle Γ_1 . The lines EI, CE, BF, FG enclose an incircle Γ_2 . The lines EJ, CE, AF, FG enclose an incircle Γ_3 . Prove that the lines BE, EI, BF, FH enclose an incircle.

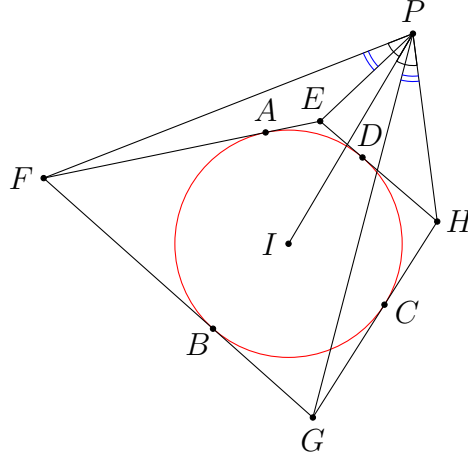


Solution. As in the proof of problem 26, by considering the circles Γ and Γ_1 , and the groups of concurrent lines EB, EJ, EC and FA, FH, FB , we see that EJ, EC, FH, FB enclose an incircle Ω_2 . Similarly, by considering the circles Γ and Γ_2 , we see that EB, EI, FA, FG enclose an incircle Ω_1 .

Let Γ_4 be the incircle touching the lines EB, EI, FH . It suffices to show that Γ_4 also touches FB . Let P be the external similitude centre of Ω_1 and Ω_2 . Applying Monge's theorem to $\Omega_1, \Omega_2, \Gamma_3$, we know that P, E, F are collinear. Similarly, applying Monge's theorem to $\Omega_1, \Omega_2, \Gamma_4$, we know that PE passes through the external similitude centre of Ω_2 and Γ_4 . As HK is an external common tangent of Ω_2 and Γ_4 , this external similitude centre is F . As BF is tangent to Ω_2 , it is also tangent to Γ_4 . This completes the proof. $\square$

Problem 28. Let $EFGH$ be a circumscribed quadrilateral with incentre I . Let P be a point outside $EFGH$ such that $\angle FPI = \angle HPI$. Prove that $\angle EPF = \angle GPH$.

Solution. Let A, B, C, D be the tangential points of the incircle with EF, FG, GH, HE respectively. Note that it suffices to prove $\angle EPI = \angle GPI$. In the following, we set up the complex plane using I as the origin and the incircle of $EFGH$ as the unit circle. We use small letters of the labels to denote the complex numbers representing the corresponding points. Note that $a\bar{a} = b\bar{b} = c\bar{c} = d\bar{d} = 1$. Also, we may assume $p = 1$.



The equations of the tangent at A and the tangent at D are

$$\begin{aligned} z + a^2 \bar{z} &= 2a, \\ z + d^2 \bar{z} &= 2d. \end{aligned}$$

Note that e satisfies both equations. Solving for e , we obtain $e = \frac{2ad^2 - 2da^2}{d^2 - a^2} = \frac{2ad}{a + d}$. Similarly, we have

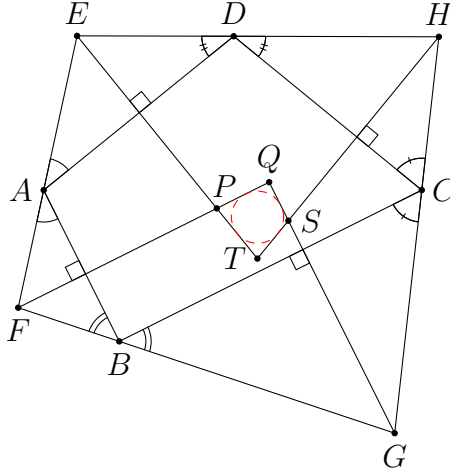
$$f = \frac{2ab}{a + b}, \quad g = \frac{2bc}{b + c}, \quad h = \frac{2cd}{c + d}.$$

Now, $\angle FPI = \angle HPI$ means PI is the angle bisector of $\angle FPH$. Therefore, the reflection of H in the line PI , which is the real axis, lies on FP . In other words, the points represented by $1, f, \bar{h}$ are collinear. Thus, $\frac{f-1}{\bar{h}-1} = \frac{\bar{f}-1}{h-1}$. Note that

$$\begin{aligned} \frac{f-1}{\bar{h}-1} &= \frac{\bar{f}-1}{h-1} \\ \Leftrightarrow \frac{(2ab - a - b)(c + d)}{(2 - c - d)(a + b)} &= \frac{(2 - a - b)(c + d)}{(2cd - c - d)(a + b)} \\ \Leftrightarrow (2ab - a - b)(2cd - c - d) &= (2 - a - b)(2 - c - d) \\ \Leftrightarrow 4abcd - 2 \sum abc &= 4 - 2 \sum a. \end{aligned}$$

By symmetry, PI is the angle bisector of $\angle EPG$ if and only if the same relation holds. This proves the desired result. $\square$

Problem 29. Let $ABCD$ be a quadrilateral. The external angle bisectors of $\angle A, \angle B, \angle C, \angle D$ form a quadrilateral $EFGH$, such that A, B, C, D lie on EF, FG, GH, HE respectively. The perpendicular lines from E, F, G, H to DA, AB, BC, CD respectively form a quadrilateral $PQST$. Prove that $PQST$ is a circumscribed quadrilateral.



Solution. Firstly, since

$$\begin{aligned}
 \angle E + \angle G &= (180^\circ - \angle EAD - \angle EDA) + (180^\circ - \angle GBC - \angle GCB) \\
 &= 360^\circ - \frac{180^\circ - A}{2} - \frac{180^\circ - D}{2} - \frac{180^\circ - B}{2} - \frac{180^\circ - C}{2} \\
 &= \frac{A + B + C + D}{2} = 180^\circ,
 \end{aligned}$$

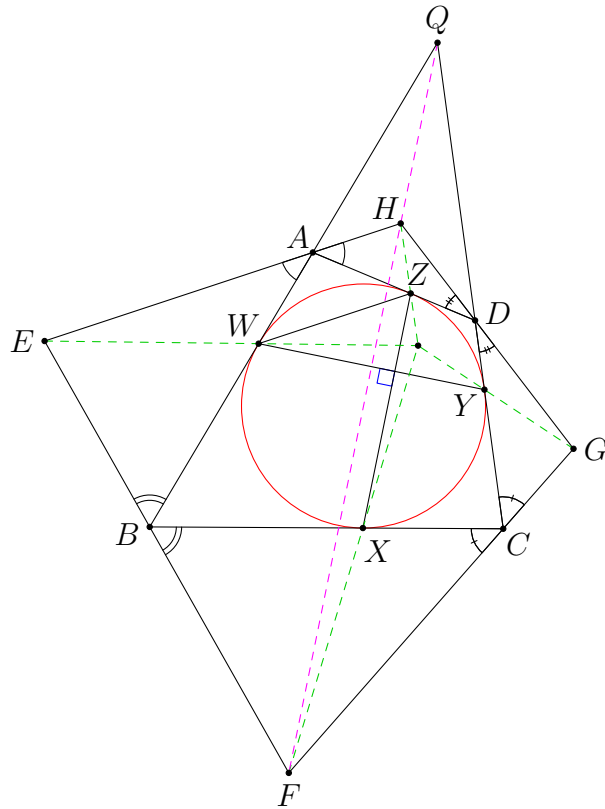
the points E, F, G, H are concyclic. As $\angle TEH = 90^\circ - \angle EDA = 90^\circ - \angle HDC = \angle THE$, we have $TE = TH$. Therefore, the internal angle bisector of $\angle T$ is the perpendicular bisector of HE . Similarly, the internal angle bisectors of $\angle P, \angle Q, \angle S, \angle T$ are the perpendicular bisectors of EF, FG, GH, HE respectively. They intersect at the centre of $(EFGH)$. So $PQST$ is a circumscribed quadrilateral. $\square$

Problem 30. Let $ABCD$ be a circumscribed quadrilateral. Let W, X, Y, Z be the tangential points of the incircle of $ABCD$ with AB, BC, CD, DA respectively. The external angle bisectors of $\angle A, \angle B, \angle C, \angle D$ form a quadrilateral $EFGH$, such that A, B, C, D lie on HE, EF, FG, GH respectively. If $WY \perp XZ$, prove that EW, FX, GY, HZ are concurrent.

Solution. Firstly, note that WZ is perpendicular to the internal angle bisector of $\angle BAD$. Thus, $WZ \parallel EH$. Similarly, $WX \parallel EF$, $XY \parallel FG$ and $YZ \parallel GH$.

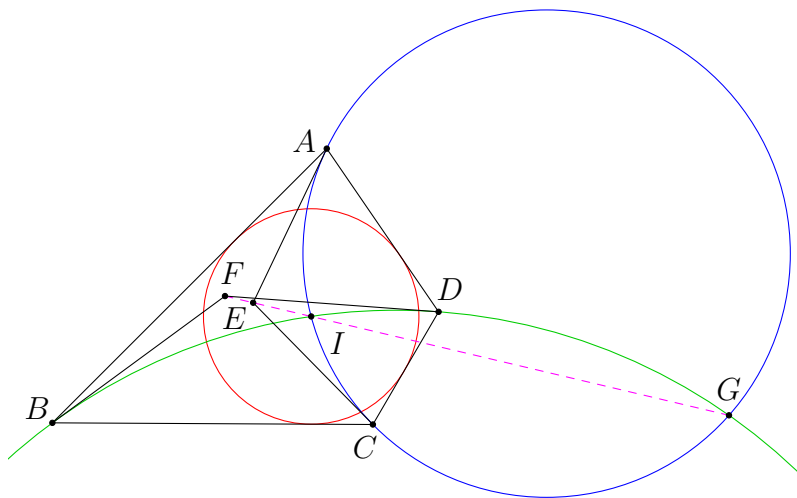
Secondly, let Q be the intersection point of AB and CD . Then H is the incentre of $\triangle QAD$, while F is the excentre of $\triangle QBC$. Therefore, QHF is the internal angle bisector of $\angle AQD$. It follows that $HF \perp WY$, and hence $HF \parallel ZX$. Similarly, $EG \parallel WY$.

Combining all these, we see that there is a homothety mapping $EFGH$ to $WXYZ$. Thus, the line joining the corresponding vertices are concurrent. $\square$

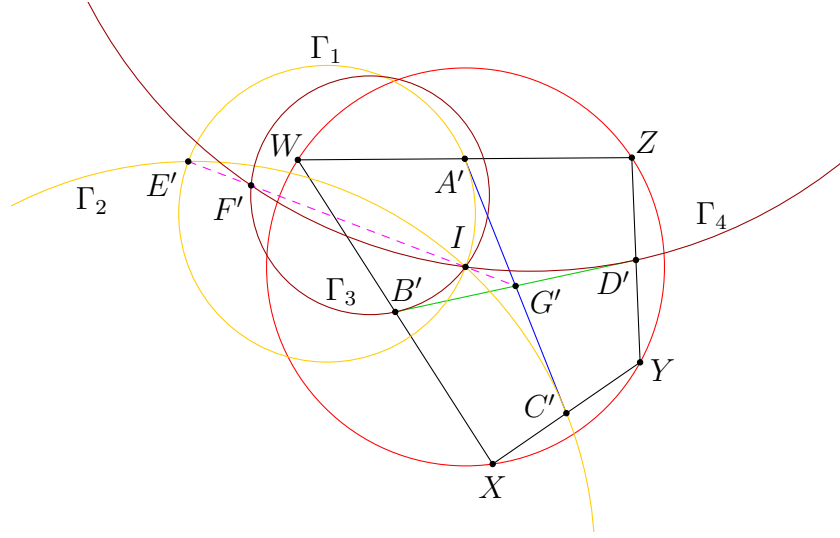


Problem 32. See Problem Set 12.

Problem 33. Let $ABCD$ be a circumscribed quadrilateral with incentre I . The tangents at A and C to (ACI) meet at E . The tangents at B and D to (BDI) meet at F . The circles (ACI) and (BDI) meet again at G . Prove that F, E, I, G are collinear.



Solution. Consider the inversion w.r.t. the incircle of $ABCD$. Let A' be the image of A , etc. Then A', B', C', D' are the midpoint of ZW, WX, WY, YZ respectively, where W, X, Y, Z are points on the incircle. The circle (AIC) is mapped to the line $A'C'$, while the circle (BID) is mapped to the line $B'D'$. Thus, G' is the intersection point of $A'C'$ and $B'D'$. By the midpoint theorem, it is easy to show that $A'B'C'D'$ is a parallelogram. So G' is the midpoint of $A'C'$ and the midpoint of $B'D'$.



Now, the tangent at A to (AIC) is mapped to a circle Γ_1 passing through A' and I and tangent to $A'C'$ at A' . The tangent at C to (AIC) is mapped to a circle Γ_2 passing through C' and I and tangent to $A'C'$ at C' . So E' is the intersection of Γ_1 and Γ_2 . As $G'A' = G'C'$, the powers of G w.r.t. Γ_1 and Γ_2 are the same. So G' lies on the radical axis $E'I$ of Γ_1 and Γ_2 . Similarly, G' lies on $F'I$. So E', F', I, G' are collinear. This implies E, F, I, G are collinear. $\square$